Supplementary Note 3. Extended details for consumables footprint calculations

The consumables footprint accounts for the cradle-to-gate upstream emissions associated with selected short-lived materials consumed in urban areas, including paper, cardboard, plastic, metal, glass, and diapers. Given the lack of detailed consumption data, waste generation serves as a proxy for urban material consumption. This method aligns with the solid waste Life Cycle Assessment (LCA) approach, which presumes that most consumables, particularly short-term items, ultimately enter the waste stream (Kissinger et al., 2013). The following sections detail the procedures used to assess material quantities and calculate their associated emissions.

PART 1: Assessing consumed material quantities

Calculating landfilled consumable quantities

First, landfilled waste quantities of each city were obtained from the Local Authorities Waste and Recycling Survey (CBS, 2022). Waste composition data were taken from the National Waste Survey (MOEP, 2014), which provides detailed information on the composition of urban waste across cities. The amounts of specific consumables in each city were estimated by multiplying each city's total generated waste (landfilled and recycled) by the percentage share of each material type in the waste composition of each city.

Calculating recycled consumable quantities

Recycled quantities of consumables for each city were also obtained from CBS (2022). Exact data were available for 15 cities, representing 62% of the study population. For the remaining cities, where detailed data were not disclosed due to confidentiality, aggregate district-level data were used. aggregated data at the district level were used. District-level per capita recycling rates for each consumable were estimated and then multiplied by the population size of each city, resulting with the recycled quantities of each consumable. Note that in order to avoid overestimation and ensure accuracy, the 15 cities with detailed recycling data were excluded from the district-level averages estimations.

PART 2: Calculating GHG emissions

For each consumable type (i), the following steps were performed:

1. **GHG emissions from landfilled consumables:** Using the landfilled quantities from Part 1, emissions were calculated by multiplying each material's landfilled amount by its respective cradle-to-gate LCA emission factor (DEFRA, 2023; Cordella et al., 2015; Kissinger et al., 2013).

$$GHG_{\text{landfilled},i} = Q_{\text{landfilled},i} \times EF_{\text{LCA},i}$$

2. **Potential GHG emissions avoided by recycling**: Using the recycled quantities from Part 1, the avoided GHG emissions were estimated by multiplying the recycled amount of each consumable by its cradle-to-gate LCA factor. This calculation reflects the potential GHG emissions that would have occurred if the material had instead been produced from virgin sources.

$$GHG_{\text{potential},i} = Q_{\text{recycled},i} \times EF_{\text{LCA},i}$$

3. **Avoided emissions from recycling**:

   - **Calculating custom net emission Factors**: For each consumable, a custom net emission factor was calculated by taking the LCA factor for producing the material from virgin sources and subtracting it with the gross emission factor for recycling that material, taken from Turner et al. (2015). The result is the custom net factor, represents the emissions saving achieved trough recycling.

$$EF_{\text{net},i} = EF_{\text{LCA},i} - EF_{\text{gross},i}$$

   - This factor represents the difference in emissions between producing a material from virgin sources and recycling that material, capturing the emissions savings achieved through recycling. If the resulting value is negative, it indicates a complete offset of emissions, and it is treated as an absolute value in calculations.
   - **Calculating avoided emissions**: Once the custom net emission factor was obtained, it was multiplied by the recycled amount of each consumable, yielding the total avoided emissions (AE) due to recycling for that material.

$$AE_i = Q_{\text{recycled},i} \times EF_{\text{net},i}$$

4. **Net carbon footprint per consumable**: The avoided emissions are subtracted from the sum of GHG emissions from landfilled quantities and potential emissions from recycling, yielding the net carbon footprint for each consumable.

$$GHG_{\text{net},i} = (GHG_{\text{landfilled},i} + GHG_{\text{potential},i}) - AE_i$$

5. **Total net carbon footprint:** Finally, the net carbon footprints of all consumables were summed to obtain the city-level total net carbon footprint.

$$GHG_{\text{total,net}} = \sum GHG_{\text{net},i}$$

The calculated emission factors are presented in Supplementary Table 6.

Supplementary Table 5. Emission Factors for Public transport by Vehicle Type, Route Type, and Traffic Conditions (tCO_2eq/km)

Vehicle Type	**Route Types**	**Traffic Situation**	**Direct EF** (tCO_2eq/km)	**Upstream EF** (tCO_2eq/km)
Bus	Innercity	Stop & Go	0.00173581	0.00029424
	Intercity-regional	Heavy	0.00093798	0.00029424
Minibus	Innercity	Stop & Go	0.00035617	0.00005844
	Intercity-regional	Heavy	0.00020490	0.00005844

Supplementary Table 6. Emission Factors of Consumables (tCO_2eq/ton)

Material	**LCA EF** (tCO_2eq/ton)	**Source**	**Gross EF** (tCO_2eq/ton)	**Custom net EF** (tCO_2eq/ton)
Paper	0.910	DEFRA, 2023	1.576	$\mid -0.666 \mid$
Cardboard	0.802	DEFRA, 2023	0.559	0.243
Plastic	3.102	DEFRA, 2023	0.339	2.763
Metal	5.255	DEFRA, 2023	0.883	4.372
Glass	1.403	DEFRA, 2023	0.395	1.008
Diapers	2.296	Adapted from Cordella et al., 2015	0.053	2.243

Supplementary Table 7. Emission Factors of Wastewater Treatment Plants (WWTPs) (tCO_2eq/m^3)

WTTP name	**Treatment type**	**Direct EF** (tCO_2eq/m^3)	**Energy-related EF** (tCO_2eq/m^3)
Afula	Enhanced (MBT)	0.000791	0.000560788
Akko	Enhanced (MBT)	0.001474	0.000560788
Al Hamra	Low	0.007925	0.000302479
Arad	Tertiary	0.001808	0.000619494
Arei HaEmeq - Yoqne'am Tiv'on	Intermediate	0.002837	0.000302479
Aruar (Ar'ara-BaNegev)	Tertiary	0.000893	0.000619494
Ashdod	Enhanced (MBT)	0.001954	0.000560788
Ashkelon	Enhanced (MBT)	0.000986	0.000560788
Ayalon (Nesher)	Enhanced (MBT)	0.000928	0.000560788
Be'er Sheva	Tertiary	0.000938	0.000619494
Bet She'an	Enhanced (MBT)	0.001580	0.000560788
Bitanya (Alummot)	Enhanced (MBT)	0.002253	0.000560788

Continued on next page...